

Salavat Galimov

Raleigh, NC | (910) 939-0267 | salgalia@proton.me | trade42.github.io | US citizen

Available: Spring and Summer 2027, in person in San Francisco; can start December 2026

Flight videos and brace CAD: <https://drive.google.com/drive/folders/158aF8HUKg1S8hgNXXOVQKNSS-r63nkcE>

EE student who builds flight and embedded electronics end to end: parts, power, wiring, bring-up, and flight test.

EDUCATION

NC State University, Raleigh, NC | BS Electrical Engineering, second-year (junior standing by credit hours) Expected May 2029

Coursework: signals, circuits and systems; computer systems; electricity and magnetism; multivariable calculus.

FLIGHT & EMBEDDED HARDWARE EXPERIENCE

Aerospace Club | Flight Projects Lead: Rocket, UAV, and RC Plane Teams 2021 – 2023

- Placed **2nd** at Aerospace Russian Nationals (Camsat Russia, **130+ teams**) with a **two-stage solid-fuel rocket** that reached ~5,000 m apogee; designed and built its **avionics**.
- Selected the parts, wired a **Raspberry Pi flight computer** to the IMU, **altimeter**, and **stabilization sensors**, and wrote the **flight software** that read the sensors and sent telemetry.
- Led the UAV team to **3rd** at nationals, flying **autonomous missions** on a **Raspberry Pi** onboard computer.
- Placed **4th** at nationals with a **1.75 m** aerobatic RC plane built from scratch (foam airframe, **carbon-fiber spars**, **brushless motors**, **LiPo** power); **4 design cycles**, **20+ test flights**.

Drop Foot Active Brace | Wearable Robotic Assistive Device (4-person team) 2026 – Present

- Delivered reliable active foot lift for an adult client with drop foot: **3 prototypes** to on-body testing in **under 30 days**.
- Selected and integrated the electronics on an **Arduino Nano** (thigh-mounted **FSR** gait sensor, dual **40 kg-cm servos** on a cable drive, **3S Li-ion** power); wrote closed-loop **C/C++ firmware** that calibrates the FSR and commands the servos.
- Fixed a **5 V rail brownout** by moving the servos onto their **own regulator**, sharing ground with the Nano, so servo current draw no longer pulled down the logic supply.
- Ran **6+ debug iterations** across power, sensing, and actuation; tested a **12 V DC motor** on an **L298N H-bridge**.

Robotics Design Challenge: Lab Automation Cell | Anthrogen (YC S24), Final Interview Round 2026

- Designed a full lab automation cell in **Fusion 360**: instrument stations, per-station robots, and sample transport.

Stealth Startup | AI Topology Optimization for UAV Airframes (Founder) 2026 – Present

- Building an AI platform that turns rough sketches into weight-optimized UAV airframes and exports them straight to **SolidWorks**; a **GNN surrogate** for finite element analysis scores stress on **1,000+ candidates**.

SOFTWARE PROJECTS

Vale | Stealth Prop Trading Firm (Founder) 2026 – Present

- Built a **Rust** research engine with **8,800+ tests** behind a **5-lane CI gate**; strategies must clear a statistical validation gate that rejects results explainable by chance, then a sealed out-of-sample holdout.
- Live tick-data pipeline across **25 crypto instruments**; every research trial is logged to a **hash-chained audit ledger**.

Vext | Real-Time Multi-AI Platform (solo build) 2026

- Async **Rust (Axum, Tokio)** backend streaming tokens to a **Next.js** frontend over **WebSockets**; **47+ Rust tests**.

LEADERSHIP & COMPETITIONS

Technology Student Association (TSA) | UAV/Drone Team Lead; VEX Robotics Captain 2023 – 2025

- Led the Drone/UAV team to **1st at state** and **7th nationally**: airframe design, build, flight tuning, and mission prep. Captained the chapter's VEX Robotics team to **1st at state**.

TECHNICAL SKILLS & AWARDS

Electronics & Power: schematic capture in Altium; Li-ion and LiPo battery power; separate logic and actuator regulation; motor drives (L298N H-bridge, servos, brushless motors); sensor integration (IMU, altitude, FSR); wiring and soldering.

Test & Debug: bench power supplies; multimeter checks (voltage, continuity); bench and flight testing; power debugging.

Embedded & Software: Arduino (C/C++), Raspberry Pi (Linux), Rust, Python, TypeScript / React / Next.js, Git; AI-assisted engineering with Claude Code.

CAD & Fabrication: SolidWorks, Fusion 360, OnShape, Blender; 3D printing; foam and carbon-fiber airframe fabrication.

Awards: Game Changers Hackathon **1st place (\$500, team captain)**; TSA **state champion** and **7th nationally**.